



# Agent Builder: Providing Relevant Context for Data Driven Agents



# Mind the AI gap

Everyone wants AI ...

... but AI projects are failing

92%

of C-suite executives plan to invest in or have already invested in AI<sup>1</sup>

#1 use case

automation of manual processes and workflows

95%

of enterprise AI solutions fail<sup>2</sup>

2x+ the rate

of other IT projects<sup>3</sup>

<sup>1</sup> [Elastic 5 insights from C-suite leaders](#)

<sup>2</sup> [MIT The GenAI Divide State Of AI in Business 2025](#)

<sup>3</sup> [RAND The Root Causes of Failure for AI Projects](#)

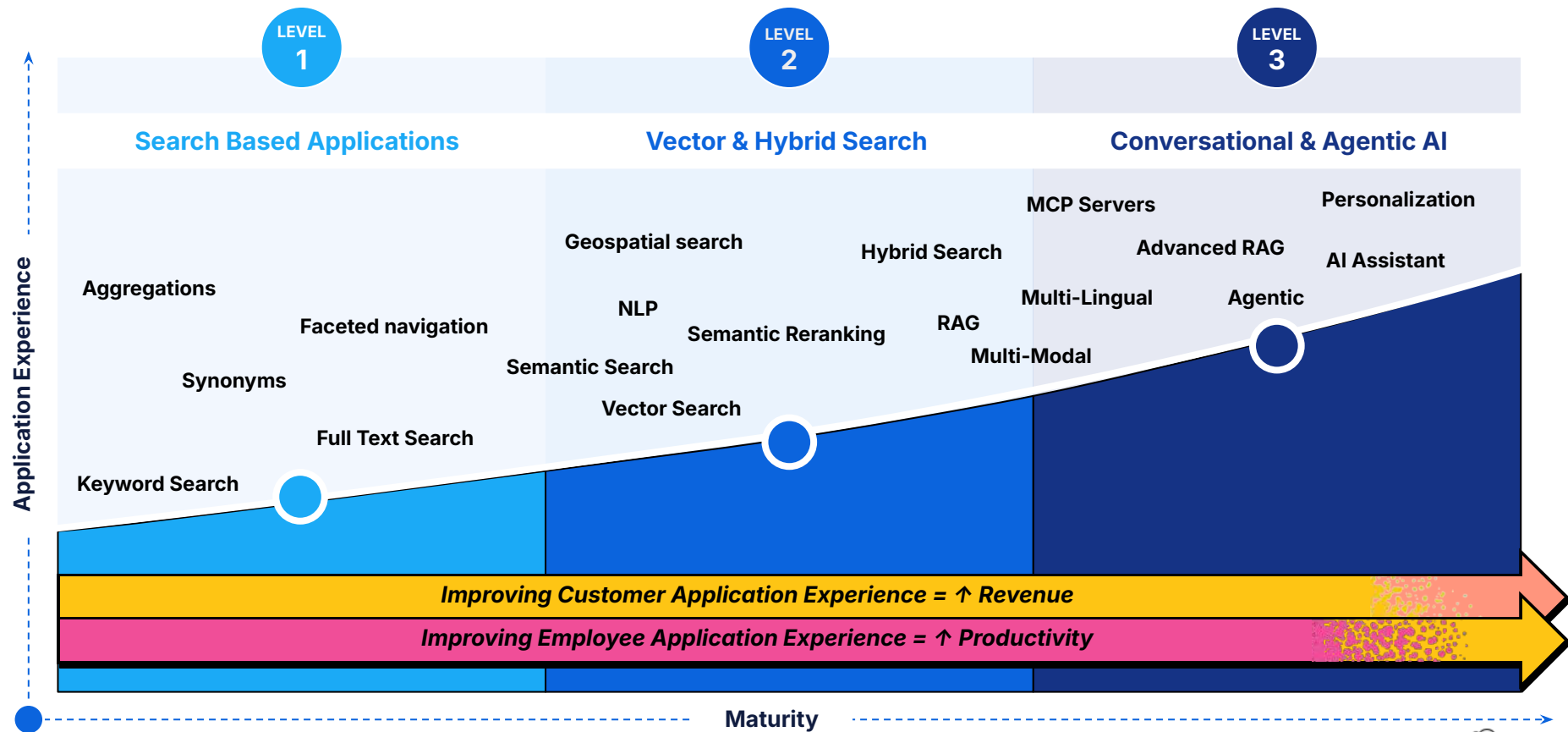


## Ashish Tiwari

Principal Solutions Architect - Search Specialist

- *10+ years in Tech*
- *Techpreneur / SMTP nerd*
- *Search / Relevance Engineer*
- *GenAI Evangelist — RAG, Agents, Vector Search*

# Journeys ...



# What are Agents?

An AI agent is a software program that can autonomously perform complex tasks by making decisions, learning from its environment, and using tools to achieve goals



# Mind the AI gap

Everyone wants AI ...

... but AI projects are failing

92%

of C-suite executives plan to invest in or have already invested in AI<sup>1</sup>

#1 use case

automation of manual processes and workflows

95%

of enterprise AI solutions fail<sup>2</sup>

2x+ the rate

of other IT projects<sup>3</sup>

<sup>1</sup> [Elastic 5 insights from C-suite leaders](#)

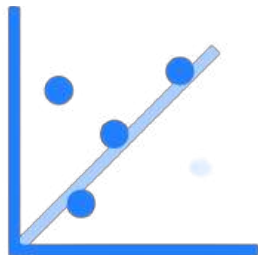
<sup>2</sup> [MIT The GenAI Divide State Of AI in Business 2025](#)

<sup>3</sup> [RAND The Root Causes of Failure for AI Projects](#)

# Why do LLMs Hallucinate?



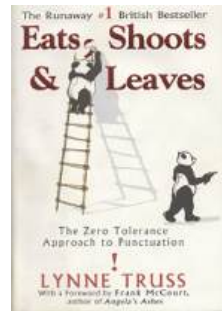
Frozen or Limited Knowledge



Model Overfitting



Dataset Biases



Language Ambiguity



# Why Language Models Hallucinate

Adam Tauman Kalai\*  
OpenAI

Ofir Nachum  
OpenAI

Santosh S. Vempala†  
Georgia Tech

Edwin Zhang  
OpenAI

September 4, 2025

## Abstract

Like students facing hard exam questions, large language models sometimes guess when uncertain, producing plausible yet incorrect statements instead of admitting uncertainty. Such “hallucinations” persist even in state-of-the-art systems and undermine trust. We argue that language models hallucinate because the training and evaluation procedures reward guessing over acknowledging uncertainty and we analyze the statistical causes of hallucinations in the modern training pipeline. Hallucinations need not be mysterious—they originate simply as errors in binary classification. If incorrect statements cannot be distinguished from facts, then hallucinations in pretrained language models will arise through natural statistical pressures. We then argue that hallucinations persist due to the way most evaluations are graded—language models are optimized to be good test-takers, and guessing when uncertain improves test performance. This “epidemic” of penalizing uncertain responses can only be addressed through a socio-technical mitigation: modifying the scoring of existing benchmarks that are misaligned but dominate leaderboards, rather than introducing additional hallucination evaluations. This change may steer the field toward more trustworthy AI systems.



# AI works best when it has **context ...**

How do I fix a red cluster?

S



A red cluster status in Elasticsearch means some data is unavailable because at least one primary shard is unassigned. Here are some steps to troubleshoot:

**Data Sources:** Elastic docs

What's wrong with my cluster and how can I fix it?

S



Your cluster is missing primary shards, which means data is unavailable and searches are failing.

I can restore them from a snapshot to recover your primary shards.

**Can I restore a snapshot?** **YES** **NO**

**Human in the loop**

**Data Sources:** Elastic docs, Product Telemetry

**Tools:** Elastic, AutoOps



## GenAI

**GenAI** excels at **rationalizing** by retrieving, synthesizing, and structuring answers **from given context**.

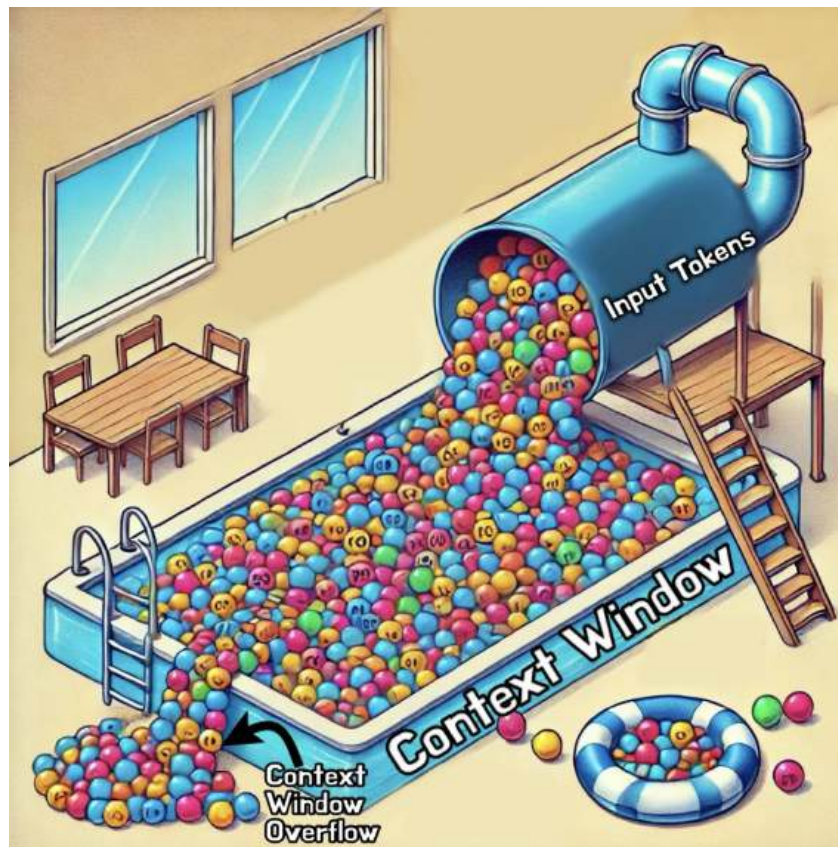


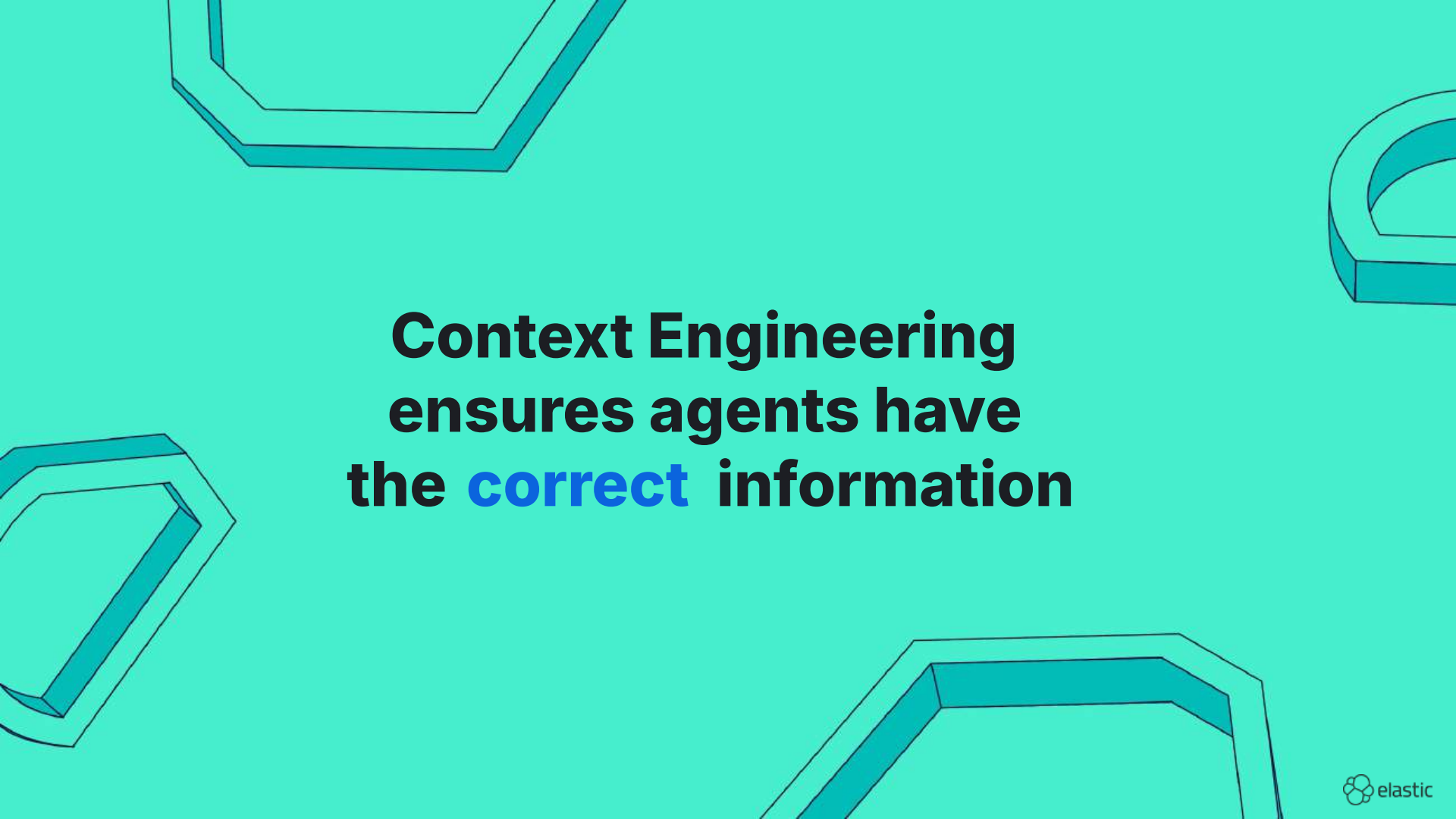
## Agentic AI

**Agentic AI** goes further, **synergizing** information and actions **from given context and tools** to execute a sequence of events **to achieve a specific goal**.

# Context Window

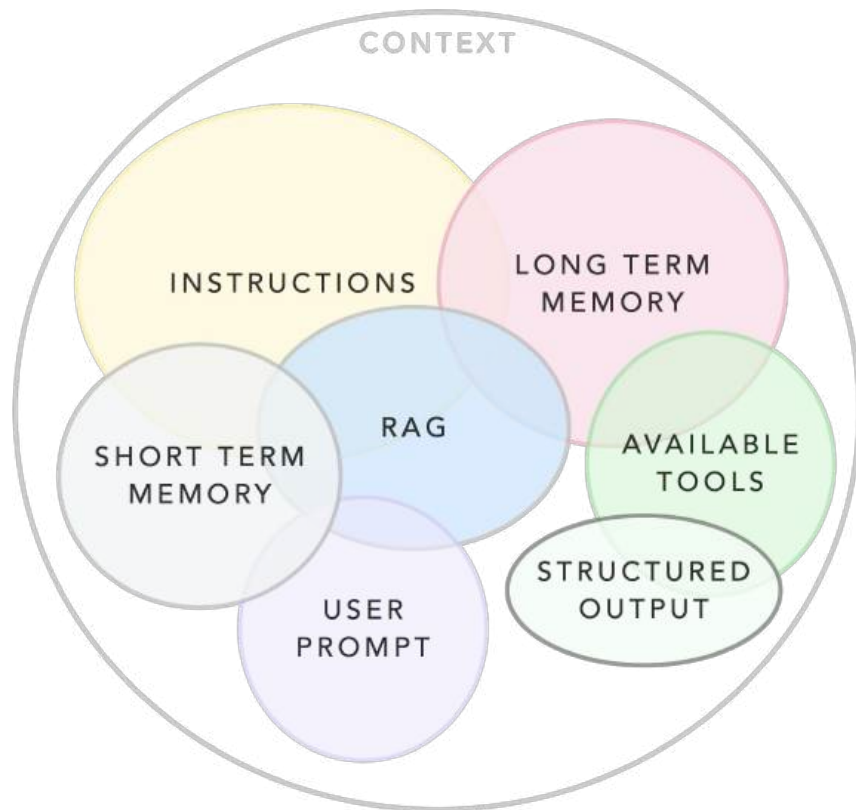
The maximum number of tokens an LLM can process at once.





**Context Engineering  
ensures agents have  
the **correct** information**

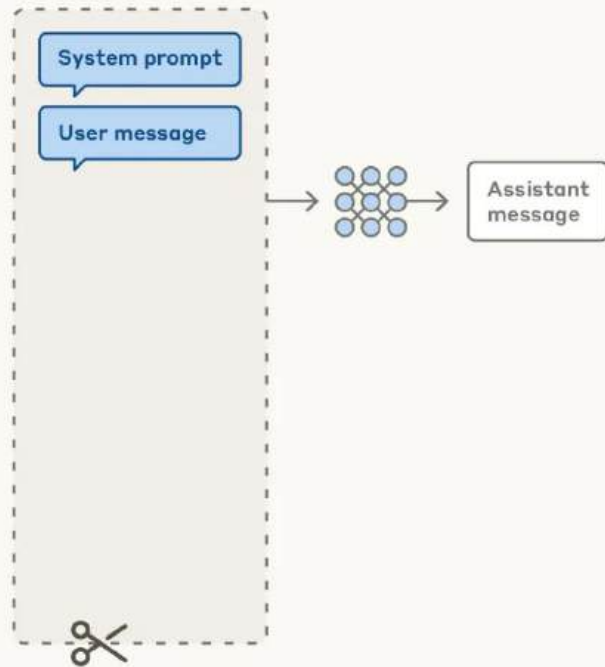
# What is Context Engineering?



# Prompt engineering vs. context engineering

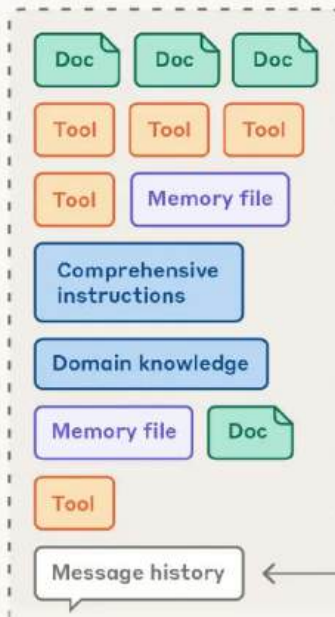
## Prompt engineering for single turn queries

Context window

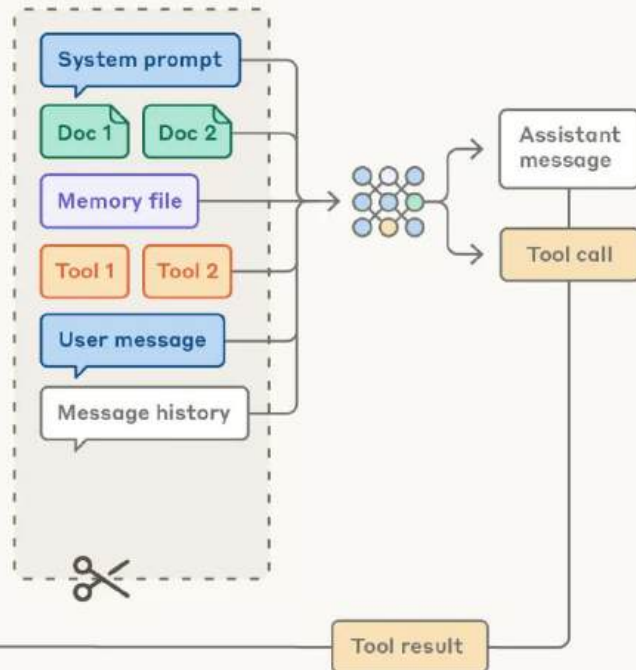


## Context engineering for agents

Possible context to give model



Context window



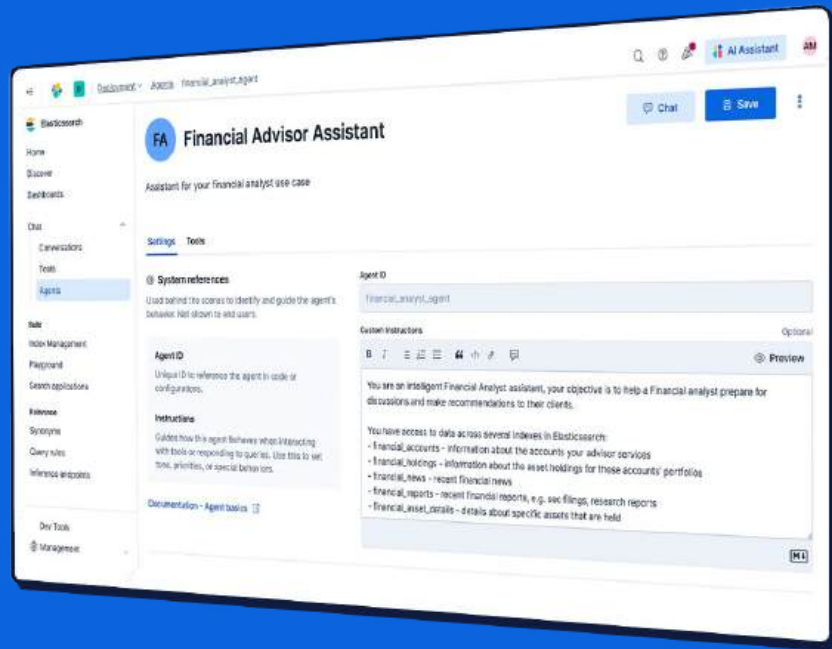
[Credit: Effective context engineering for AI agents | Anthropic](#)



**NEW**

# Generally Available **Elastic Agent Builder**

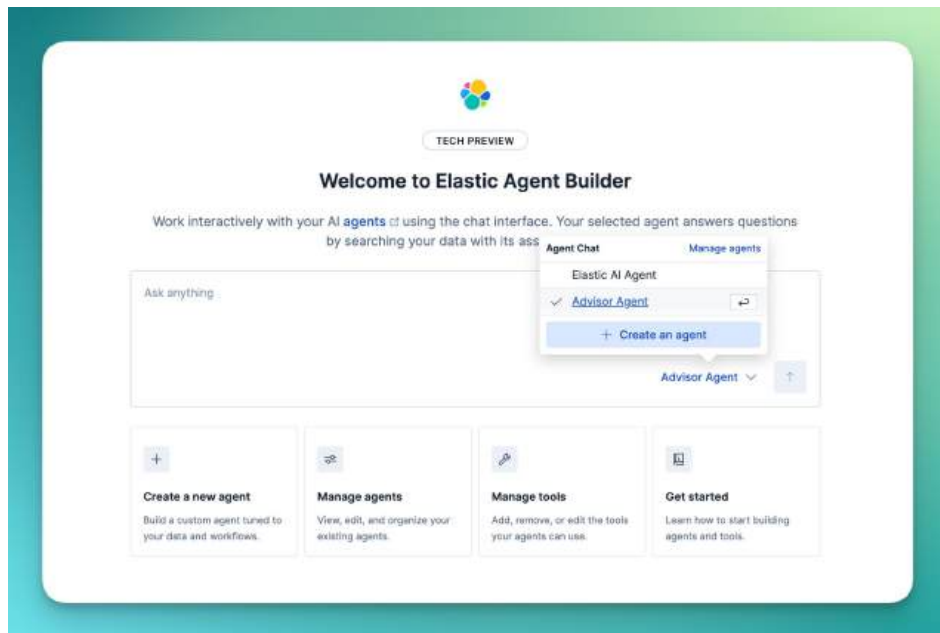
- Simplify the development of custom AI agents
- Natively chat with your Elasticsearch data



# Meet Elastic Agent Builder

Develop custom AI agents and chat experiences in minutes with Elasticsearch leading relevance and new agent-building capabilities

- **Natively chat with any data in Elastic** using a built-in agent
- **Build custom AI agents** that achieve higher accuracy, relevance, and efficiency based on the power of hybrid search
- **Create powerful tools:** Give your agent new powers. Build custom tools with the full power of ES|QL for fine-grained control over data, relevance, and security.
- **Expose** your data by hosting **MCP and A2A** compatible interfaces to your AI ecosystem.
- **Use Workflows** to accurately execute internal and external processes to take **actions & gather context**

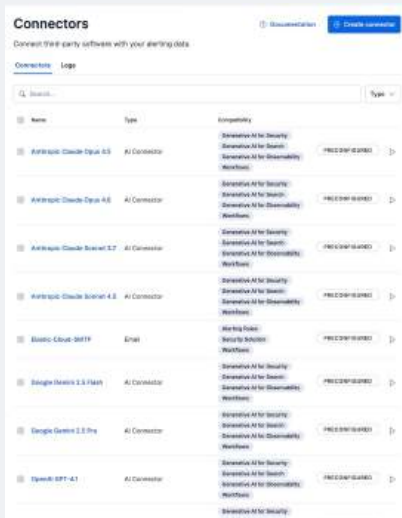




# Which LLM?

## Default

LLMs provided via Elastic Inference Service (EIS)

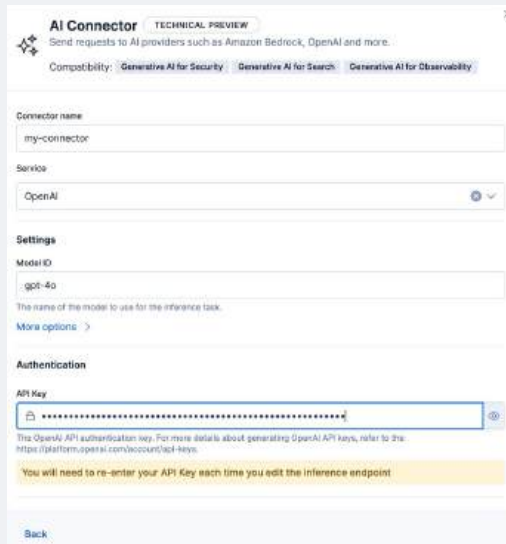


The screenshot shows the 'Connectors' page in the Elastic UI. It features a search bar and a table of connectors. The table has columns for Name, Type, and Compatibility. The connectors listed are primarily AI Connectors for various AWS services like Amazon SageMaker and Amazon Bedrock, as well as an Elastic Cloud Search connector.

| Name                           | Type         | Compatibility                                                                                          |
|--------------------------------|--------------|--------------------------------------------------------------------------------------------------------|
| Amazon SageMaker Endpoint V2   | AI Connector | Generative AI for Security<br>Generative AI for Search<br>Generative AI for Observability<br>Workflows |
| Amazon SageMaker Endpoint V1   | AI Connector | Generative AI for Security<br>Generative AI for Search<br>Generative AI for Observability<br>Workflows |
| Amazon SageMaker Endpoint V1.1 | AI Connector | Generative AI for Security<br>Generative AI for Search<br>Generative AI for Observability<br>Workflows |
| Amazon SageMaker Endpoint V1.2 | AI Connector | Generative AI for Security<br>Generative AI for Search<br>Generative AI for Observability<br>Workflows |
| Amazon SageMaker Endpoint V1.3 | AI Connector | Generative AI for Security<br>Generative AI for Search<br>Generative AI for Observability<br>Workflows |
| Elastic Cloud Search           | EI Connector | Generative AI for Security<br>Generative AI for Search<br>Generative AI for Observability<br>Workflows |
| Amazon SageMaker Endpoint V1.4 | AI Connector | Generative AI for Security<br>Generative AI for Search<br>Generative AI for Observability<br>Workflows |
| Amazon SageMaker Endpoint V1.5 | AI Connector | Generative AI for Security<br>Generative AI for Search<br>Generative AI for Observability<br>Workflows |
| Amazon SageMaker Endpoint V1.6 | AI Connector | Generative AI for Security<br>Generative AI for Search<br>Generative AI for Observability<br>Workflows |

## Connector

Create a custom connector using the AI Connector (tech preview)



The screenshot shows the 'AI Connector' configuration page. It includes a header with 'AI Connector' and 'TECHNICAL PREVIEW' labels. Below the header, there's a section for 'Connector name' with the value 'my-connector'. The 'Service' dropdown is set to 'OpenAI'. The 'Settings' section shows 'Model ID' as 'gpt-4o'. The 'Authentication' section has a text input for the 'API Key'.

**AI Connector** TECHNICAL PREVIEW  
Send requests to AI providers such as Amazon Bedrock, OpenAI and more.

Compatibility: **Generative AI for Security** **Generative AI for Search** **Generative AI for Observability**

Connector name  
my-connector

Service  
OpenAI

**Settings**  
Model ID  
gpt-4o  
The name of the model to use for the inference task.  
[More options](#)

**Authentication**  
API Key  
[Redacted API Key]

This OpenAI API authentication key. For more details about generating OpenAI API keys, refer to the <https://platform.openai.com/docs/api-keys>.

You will need to re-enter your API Key each time you edit the inference endpoint

[Back](#)

# In-built tools

There are currently 11 tools available, prefixed by

`platform.core`:

- Execute ES|QL queries
- Generate ES|QL query from natural language
- Get full document content based in ID
- Retrieve mappings for one or more specified indices
- List relevant indices, aliases and streams based on natural language query
- List indices, aliases and datastreams in the cluster
- Core search tool
- ... and more!

The screenshot displays the 'Tools' section of the Elasticsearch Agent Builder interface. The top navigation bar includes 'temp-20260218', 'Agent Builder', and 'Tools'. The left sidebar contains navigation links for 'Elasticsearch', 'Discover', 'Dashboards', 'Agents', 'Machine Learning', and 'Data management'. The main content area is titled 'Tools' and includes a search bar, a 'Labels' dropdown, and a list of 11 tools. The tools are displayed in a table with columns for 'ID', 'Description', and 'Labels'. The tools are:

- `platform.core.cases`: Retrieves cases from Elastic Security, Observability, or Stack Management. Supports three operation modes: `cases`.
- `platform.core.execute_esql`: Execute an ES|QL query and return the results in a tabular format.
- `platform.core.generate_esql`: Generate an ES|QL query from a natural language query.
- `platform.core.get_document_by_id`: Retrieve the full content (source) of an Elasticsearch document based on its ID and index name.
- `platform.core.get_index_mapping`: Retrieve mappings for the specified index or indices.
- `platform.core.get_workflow_execution_status`: Retrieve the status of a workflow execution.
- `platform.core.index_explorer`: List relevant indices, aliases and datastreams based on a natural language query.
- `platform.core.integration_knowledge`: Search and retrieve knowledge from Fleet-installed integrations. This includes information on how to configure and use integrations for data ingestion into Elasticsearch. Labels: `integration`, `knowledge-base`, `fleet`.
- `platform.core.list_indices`: List the indices, aliases and datastreams from the Elasticsearch cluster.
- `platform.core.product_documentation`: Search and retrieve documentation about Elastic products (Kibana, Elasticsearch, Elastic Security, Elastic Observability).
- `platform.core.search`: A powerful tool for searching and analyzing data within your Elasticsearch cluster.

The bottom of the interface shows 'Rows per page: 25' and a pagination control.

# Enabling Conversational & Agentic experiences

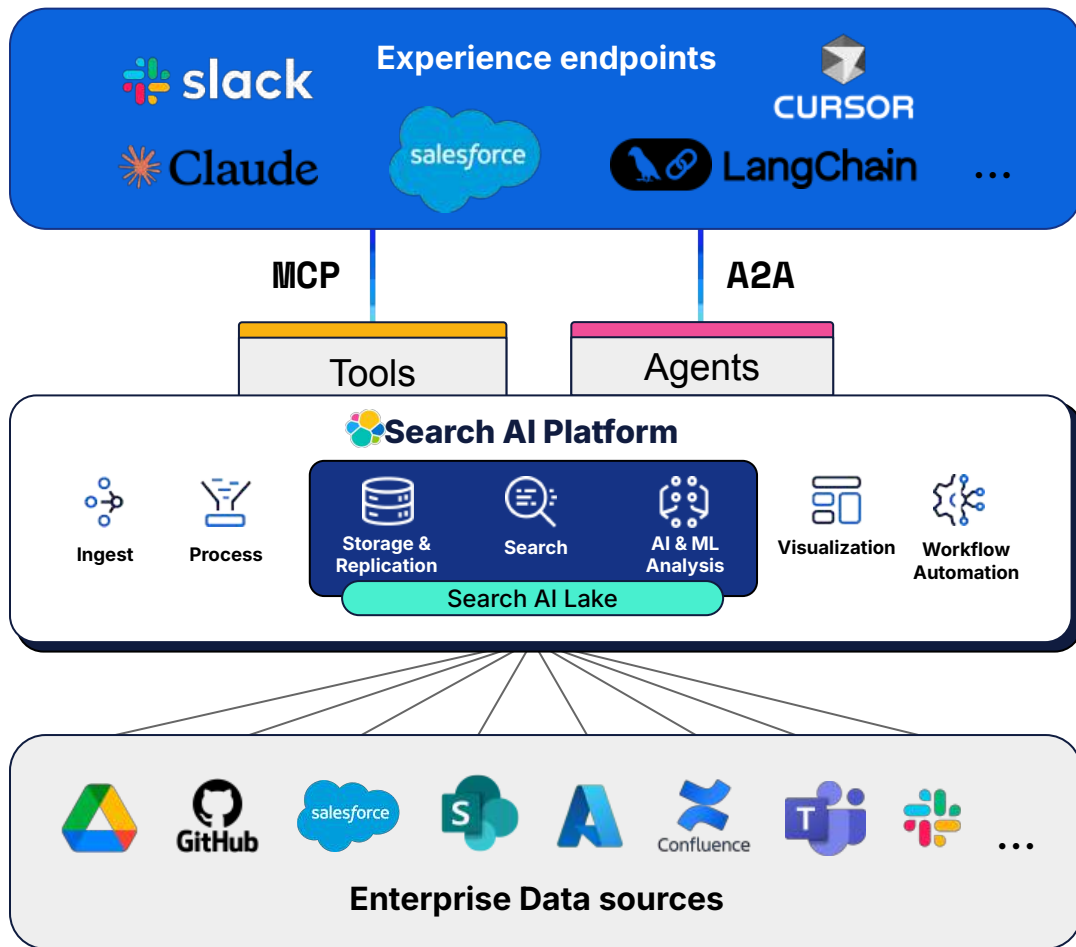
Integrate tools and  
agents into existing  
user & developer  
flows and  
experiences

AI Native  
Experiences

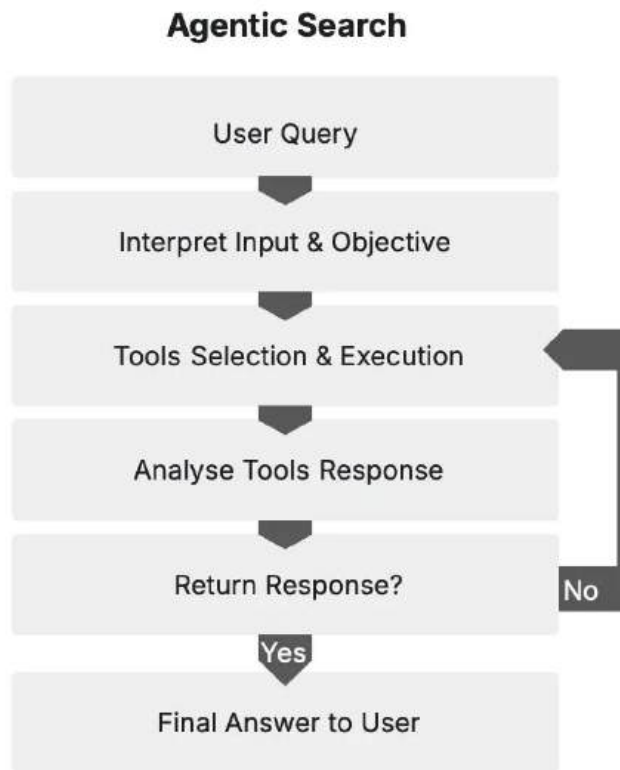
Powered by...  
Tools & Agents

Enabled by...  
The Platform

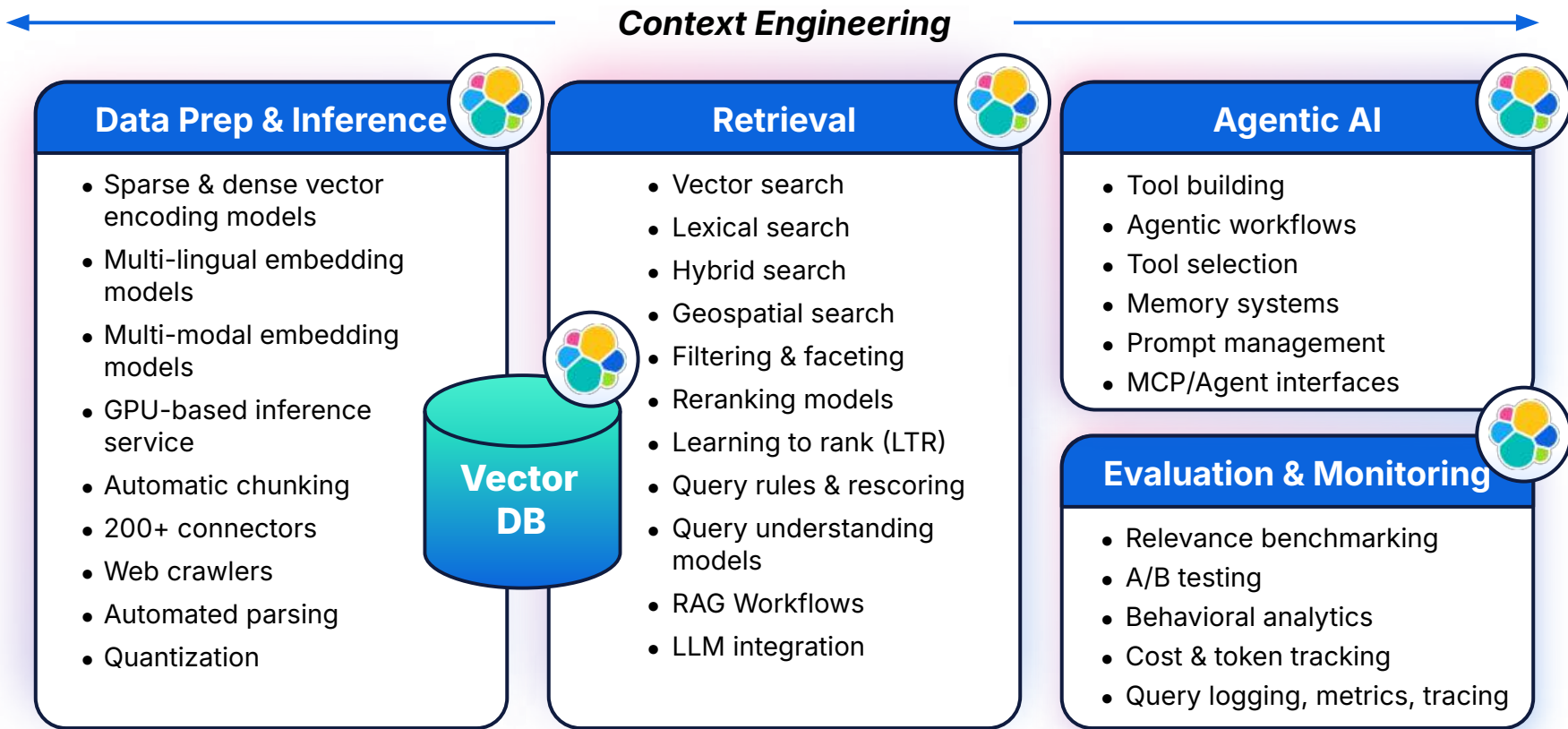
Built on...  
Enterprise Data



# How do Agents work in Elastic?

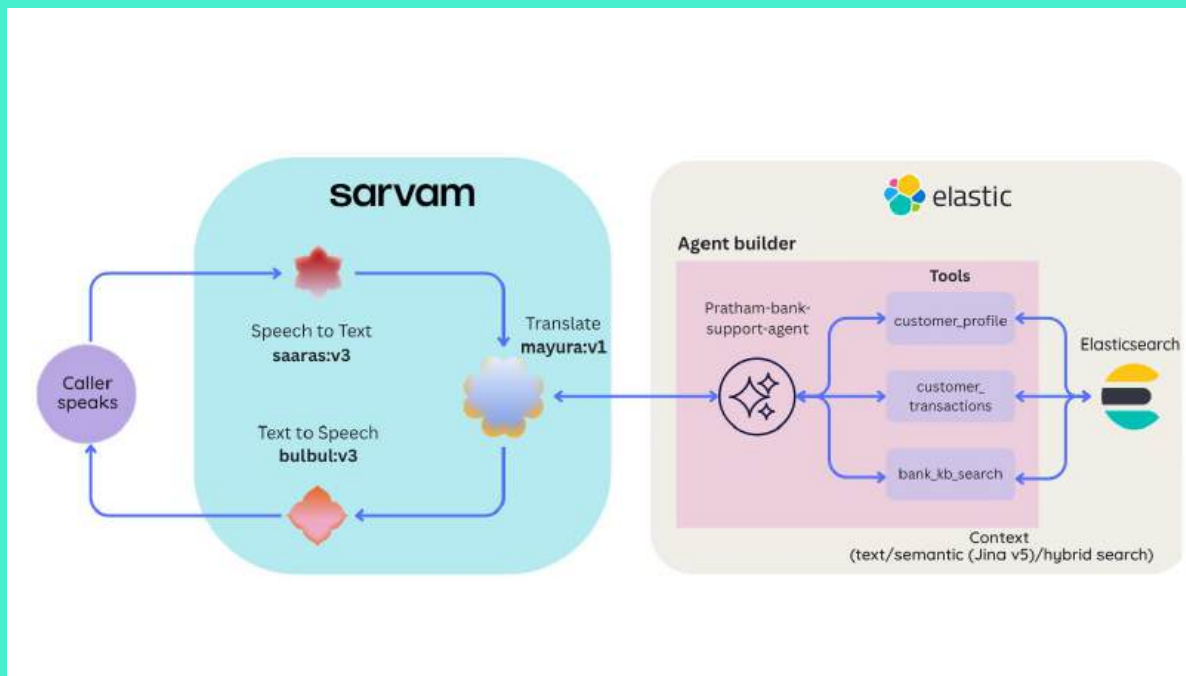


# Elastic provides everything agent need



# Demo:

## Multilingual voice agent with Elastic Agent Builder & Sarvam AI



# Demo:

## Multilingual voice agent with Elastic Agent Builder & Sarvam AI

**Pratham Bank - Mitra**  
"Where is my money?" voice support - आपका बैंकिंग साथी

Powered by  
Elastic Agent Builder + Sarvam AI

CALLER (DEMO LOGIN)

New call

**Rajesh Kumar**  
Bengaluru - A/C \*\*6789  
Salaried customer — salary just credited, home-loan EMI debited, a Flipkart refund still pending.

**Priya Sharma**  
Pune - A/C \*\*4321  
Money debited without consent — a disputed ₹14,999 UPI payment to an unknown payee (UPI fraud scenario).

**Anil Verma**  
Lucknow - A/C \*\*2345  
Pensioner — this month's pension is delayed and showing as processing ("meri pension aayi ya nahin?").

**Sunita Devi**  
Patna - A/C \*\*9765  
Waiting on a ₹2,499 Meesho return refund that is still processing.

Verification hint (for you): DOB is **14 March 1988** — Mitra will ask for it before sharing money details.

Language: **Auto** हिंदी मराठी English



Tap to speak - बोलने के लिए टैप करें

Did my salary come this month? Where is my Flipkart refund? What is my current balance?

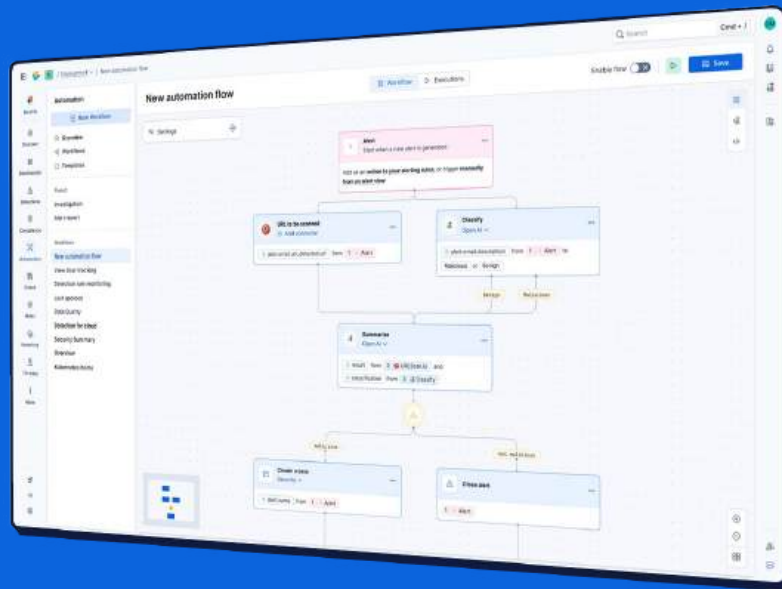
...or type your question here

Send

**NEW**

# Introducing **Elastic Workflows**

- Automate actions directly where your data lives
- Both rules-based and agentic automation

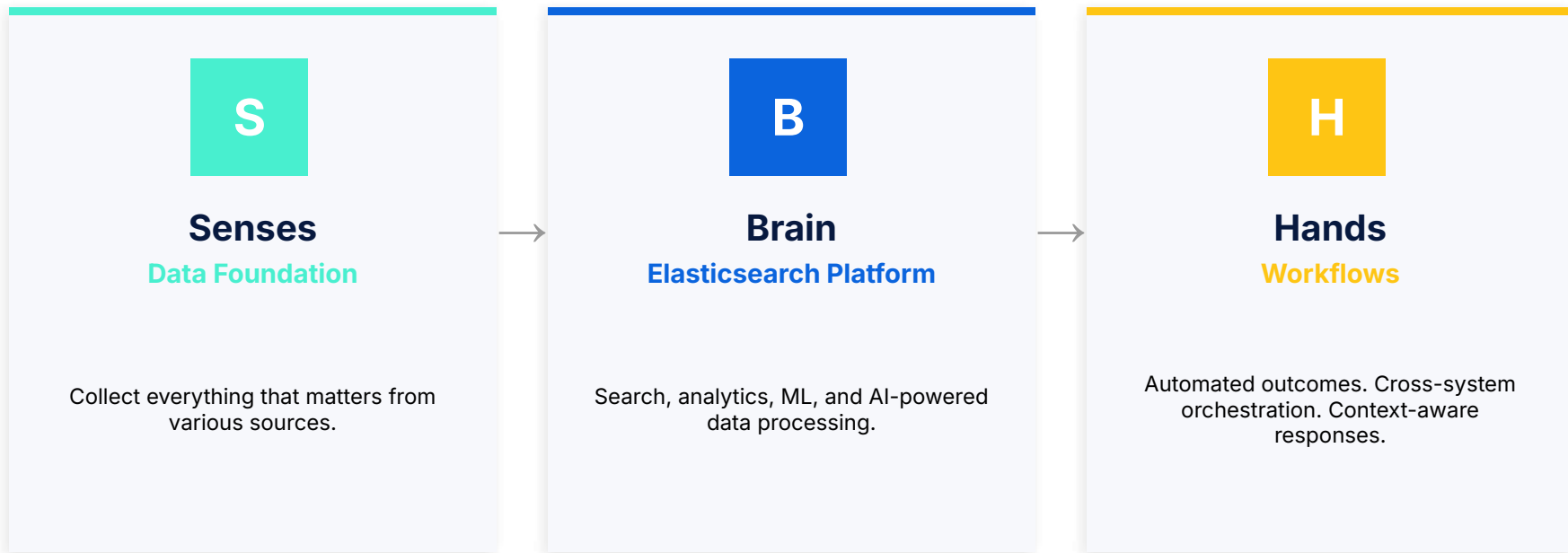




# Elastic Workflows

- Built for search teams: The native automation engine built into the Elasticsearch platform
- Define automation to accurately execute internal and external processes to take actions & gather context.
- Combine rule-based automation and AI Agents to automate routine and complex tasks
- No external tools. No external services and scripts. No integration overhead.
- Defined in YAML - simple enough to version-control, expressive enough for real logic

# Platform Architecture: Senses, Brain, and Hands



*Elasticsearch stays at the center — every workflow step has access to the full context of your environment*

# Workflow Anatomy: YAML Structure

```
name: Semantic Search & RAG
description: Retrieve context and generate AI response
tags: [search, generative-ai, rag]

consts:
  indexName: national-parks-vector
  modelId: ".elser_model_2"

inputs:
  userQuery:
    type: string
    description: Question from the end-user

triggers:
  - type: manual

steps:
  - id: semantic_retrieval
    type: elasticsearch.search
    index: "{{ consts.indexName }}"
    body:
      query:
        text_expansion:
          ml.tokens:
            model_id: "{{ consts.modelId }}"
            model_text: "{{ inputs.userQuery }}"
```

## ● name / description / tags

Metadata for identification and search

## ● consts

Fixed values reused across the workflow. Accessed via `{{ consts.modelId }}`

## ● inputs

Parameters that vary per execution. User-provided at runtime.

## ● triggers

How the workflow starts: manual, scheduled, alert, or webhook.

## ● steps

The sequence of actions. Each step has an id, type, and configuration.

# Triggers: How Workflows Start

## Manual

Explicit user action. Click Run in Kibana or invoke via API. Best for: Testing a new RAG prompt, manual synonym injection, or ad-hoc data enrichment.

```
triggers:  
  - type: manual
```

## Scheduled

Run at intervals or cron patterns. Best for: Daily search-performance reports, weekly re-ranking of "trending" products, or periodic vector embedding updates.

```
triggers:  
  - type: scheduled  
    interval: 1h
```

## Alert

Triggered by Kibana alerting rules. Best for: Detecting "Zero Result" search spikes, identifying "low-click" queries, or monitoring AI model hallucinations.

```
triggers:  
  - type: alert  
    rule_type: .es-query
```

## Webhook

External systems send HTTP requests. Best for: Receiving feedback loop data from a custom LLM frontend. Or when an agent identifies a gap in your knowledge base, it can hit the webhook to trigger an automated document ingestion process.

```
triggers:  
  - type: webhook  
    path: /my-hook
```

# Agents can help you ...



Improve technical support resolution times with Agents to aid investigations



Improve sales reporting and analysis with Agents to summarize and deliver insights



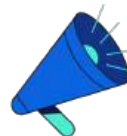
Understand and create policies (e.g. HR, IT, Legal) based on existing and usage (e.g. past searches/conversations)



Investigate purchase and product trends to make better recommendations



Create intelligent product or content recommendations based on catalog and past searches, interactions



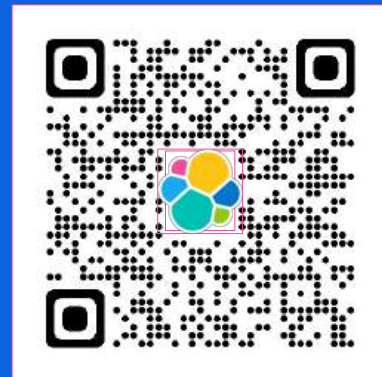
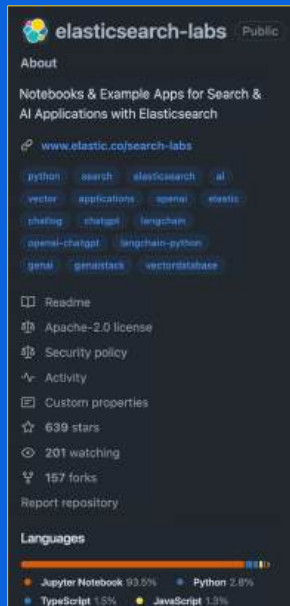
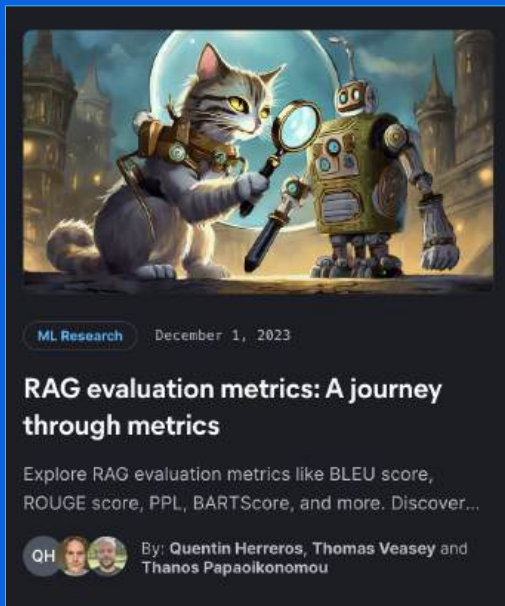
Any other internal knowledge worker use case...



Ensure compliance across unstructured/structured data sources (e.g. KYC)

RESOURCES FOR DEVELOPERS | BY DEVELOPERS LIKE YOU!

# Search Labs



- ➔ Tutorials
- ➔ GitHub Example Code
- ➔ Integrations
- ➔ Blogs

[elastic.co/search-labs](https://elastic.co/search-labs)

[github.com/elastic/elasticsearch-labs](https://github.com/elastic/elasticsearch-labs)



# Thank you!

